

BORDER OPTICS

A Satellite-Based Verification Framework for India's Vibrant Villages Programme (VVP-I)
Executive Summary · DOI: 10.5281/zenodo.21759970 · Sakshi D. Maske

The Question

Asked directly in Parliament whether the Vibrant Villages Programme's impact had ever been assessed, the Ministry of Home Affairs answered without qualification: “No impact assessment has been carried out” (Lok Sabha Unstarred Question No. 508, 3 February 2026). Did India's Rs. 4,800 crore border-village development scheme produce measurable physical development on the ground — and can the answer be trusted, or does it depend on an arbitrary measurement choice?

The Method

258 individually geocoded villages across five Himalayan border states/UTs (Arunachal Pradesh, Sikkim, Uttarakhand, Himachal Pradesh, Ladakh) were tested using Sentinel-2 built-up-area change (NDBI) and VIIRS night-lights, 2021–2025, against a matched non-VVP control group of 753 villages in the same 14 districts. Every test was run under two independent satellite compositing windows — full-year and summer-matched — specifically to test whether the methodology choice itself changes the answer, using Wilcoxon signed-rank tests for before/after change, Spearman correlations for budget and border-proximity relationships, and a district-fixed-effects Difference-in-Differences model against the control group. The core two-year comparison is extended to a three-point 2021/2023/2025 trend, and the fixed 500m extraction buffer is tested against 250m and 1km alternatives.

The Finding

Built-up area change is compositing-window sensitive, not confirmed. A full-year composite shows no significant increase ($p = 1.000$); the identical villages under a summer-matched composite show a highly significant increase ($p < 0.000001$) — traced to snow-cover contamination in the full-year window and monsoon cloud cover eliminating Sikkim's data in the summer window. That significant summer result survives three further stress tests: it is significantly larger than the change in a matched non-VVP control group (district-FE DiD, $p = 0.0021$), it holds at 250m and 1km buffer radii as well as 500m (all $p < 0.002$, sample-matched), and it clears Holm-Bonferroni correction for multiple testing. But it is not a steady trend — a three-point 2021/2023/2025 extension shows the change is concentrated in a 2023-to-2025 recovery following an earlier 2021-to-2023 decline.

Test	Full-Year Window	Summer-Matched Window
NDBI change (Wilcoxon)	$p = 1.000$ — not significant	$p < 0.000001$ — highly significant
Night-lights change (Wilcoxon)	$p = 0.050$	$p = 0.9999$
Control-group DiD (NDBI, H4)	$p = 0.215$ — not significant	$p = 0.0021$ — significant
Budget vs. measured change	~10x budget gap (AP vs. UK) → nearly identical mean change	

Night-lights, the more temporally stable proxy, shows a consistent null across both windows — carrying more weight than the window-sensitive NDBI result. Arunachal Pradesh's sanctioned budget (Rs. 2,749.74 crore) is roughly ten times Uttarakhand's (Rs. 270.58 crore), yet the two states' mean measured built-up change is nearly identical — descriptively consistent with budget-independent implementation.

Validation & Robustness Checklist

- ✓ Dual compositing-window test (full-year + summer-matched)
 - ✓ Two independent metrics tested separately (NDBI + VIIRS)
 - ✓ Matched non-VVP control group (753 villages, same 14 districts, DiD design)
 - ✓ Three-point multi-year trend (2021/2023/2025), not a single before/after pair
 - ✓ Buffer-radius sensitivity swept (250m/500m/1km)
 - ✓ Cross-checked against sanctioned budget (Parliamentary record)
 - ✓ Every data gap disclosed (Ladakh, HP, Uttarakhand caveats)
- ! NDBI result flagged as window-sensitive — and, once significant, shown to be concentrated in 2023–2025 rather than sustained since sanction

Honest Limitation

This project's own headline result is an instability disclosure rather than a clean confirmation: Himalayan terrain introduces genuine seasonal measurement artifacts (snow cover in winter months, monsoon cloud cover eliminating usable Sikkim data in summer) that make a single compositing choice unreliable evidence on its own. The control-group comparison strengthens the surviving signal but rests on a baseline NDBI-level imbalance between treated and control villages, not a confirmed shared pre-trend. The border-proximity (H3) correlation is explicitly treated as exploratory given sample size and border-geometry caveats. Data gaps — Ladakh's unresolved villages, Himachal Pradesh's partial sample, Uttarakhand's block ambiguity — are disclosed rather than smoothed over.

Real-World Relevance

This project is the independent impact assessment Parliament was told does not exist for a Rs. 4,800 crore government scheme spanning 2,967 sanctioned villages. It demonstrates that satellite-based verification of public infrastructure spending is both feasible and necessary — and that when the two most defensible measurement choices produce opposite conclusions, disclosing that instability, then stress-testing whichever result survives against a control group, a multi-year trend, and a methodological sensitivity sweep, is more useful to policymakers than a false confidence in either single result.

